

Rechnerarchitektur - Übung 6

Erweitern & Analysieren vom Multicycle Prozessor

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DI Simon Reitingner

21.05.2026

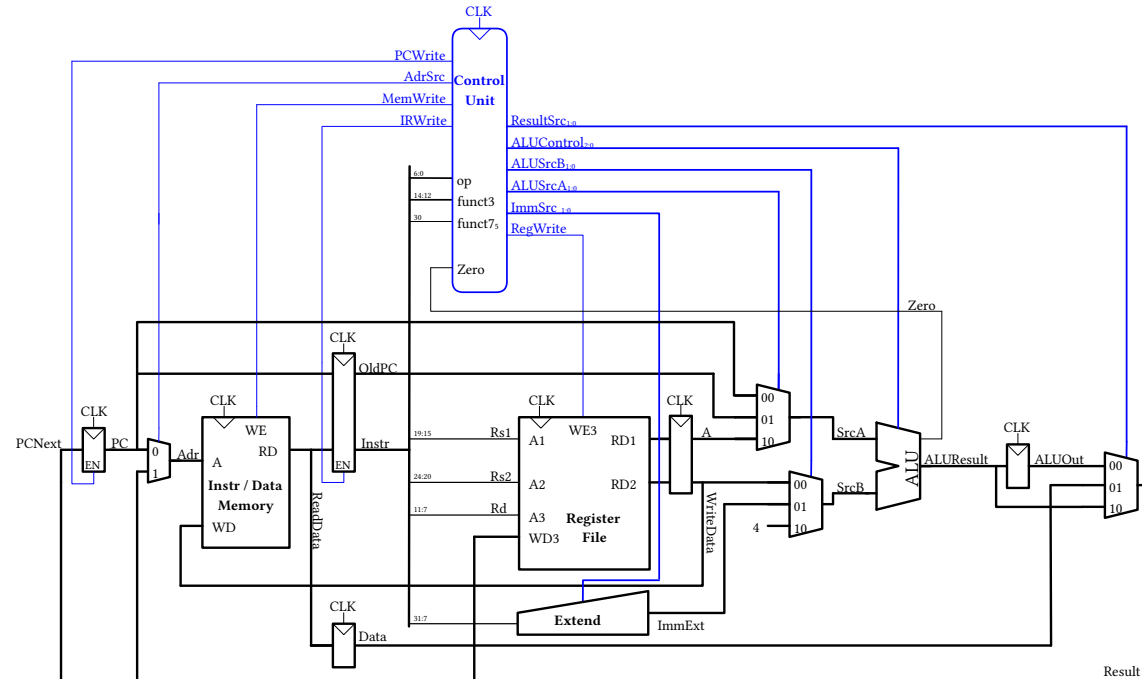
Multicycle - Erweiterung: lui

Instruktion	Name	Typ	Beschreibung
lui rd, upimm	Load Upper Imm.	U	rd = imm << 12

Wir möchten den Multicycle Prozessor um den lui Befehl erweitern.

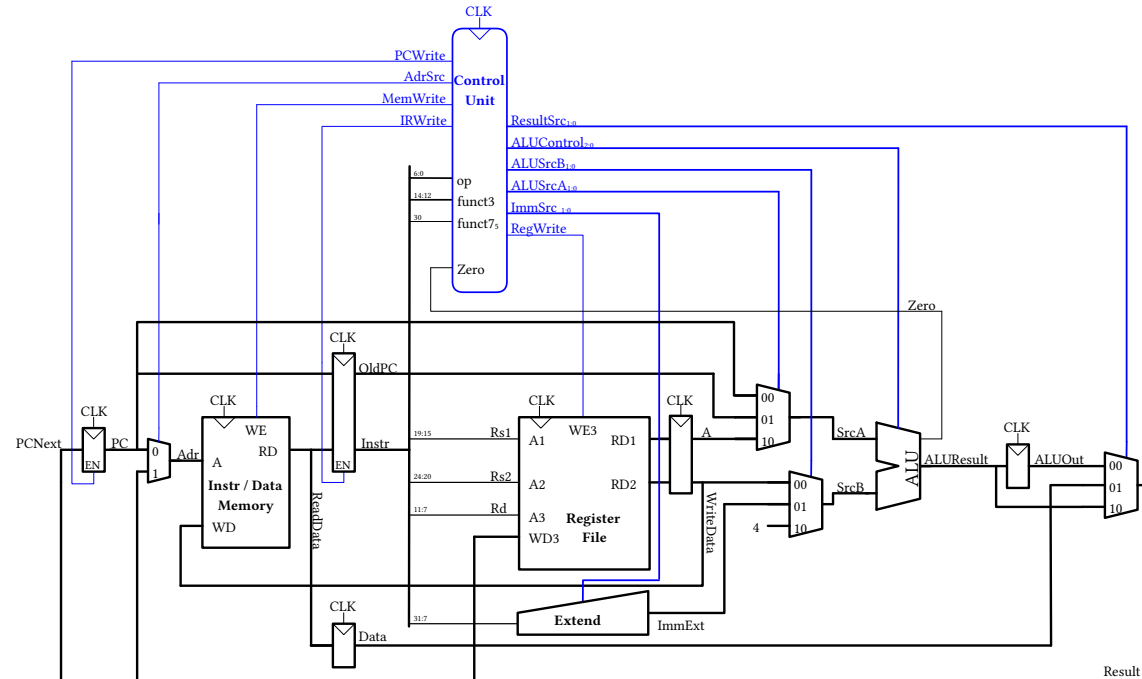
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Immediate für U Instruktionen

ImmSrc _{1:0}	ImmExt	Instr. Type
00	$20 \times \text{instr}[31], \text{instr}[31:20]$	I-Type
01	$20 \times \text{instr}[31], \text{instr}[31:25], \text{instr}[11:7]$	S-Type
10	$19 \times \text{instr}[31], \text{instr}[31], \text{instr}[7], \text{instr}[30:25], \text{instr}[11:8], 0$	B-Type
11	<i>reserved</i>	J-Type

Core Instruction Formats

31	25	24	20	19	15	14	12	11	7	6	0	
funct7		rs2		rs1		funct3		rd		opcode		R-type
imm _[11:0]				rs1		funct3		rd		opcode		I-type
imm _[11:5]		rs2		rs1		funct3		imm _[4:0]		opcode		S-type
imm _[12,10:5]		rs2		rs1		funct3		imm _[4:1,11]		opcode		B-type
imm _[31:12]								rd		opcode		U-type
imm _[20,10:1,11,19:12]								rd		opcode		J-type
7bit		5bit		5bit		3bit		5bit		7bit		

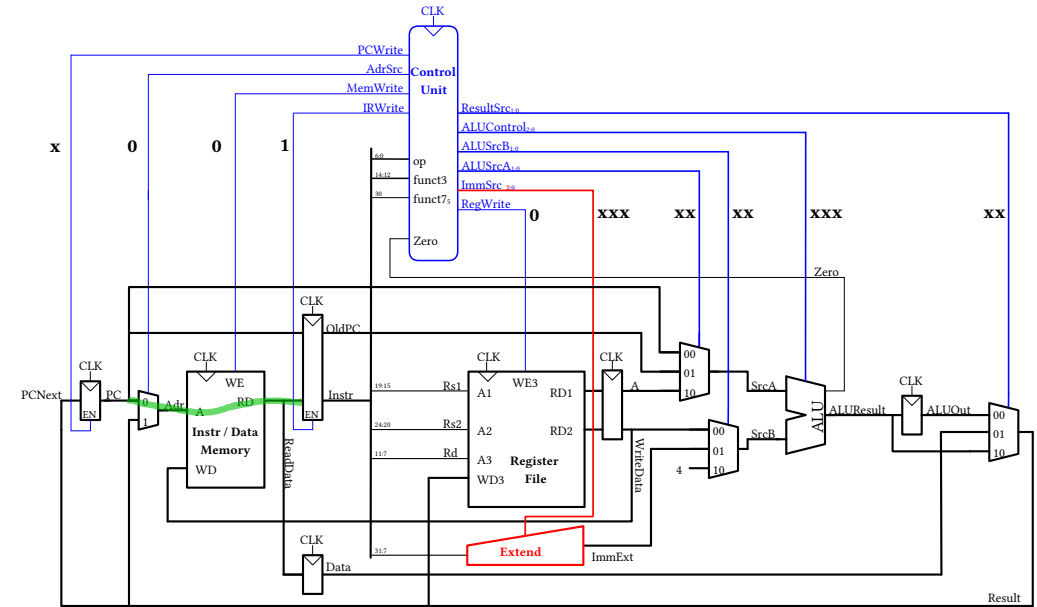
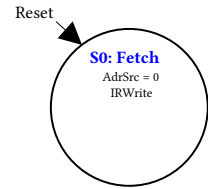
Immediate für U Instruktionen

ImmSrc _{2:0}	ImmExt	Instr. Type
000	$20 \times \text{instr}[31], \text{instr}[31:20]$	I-Type
001	$20 \times \text{instr}[31], \text{instr}[31:25], \text{instr}[11:7]$	S-Type
010	$19 \times \text{instr}[31], \text{instr}[31], \text{instr}[7], \text{instr}[30:25], \text{instr}[11:8], 0$	B-Type
011	<i>reserved</i>	J-Type
100	$\text{instr}[31:12], 12 \times '0'$	U-Type

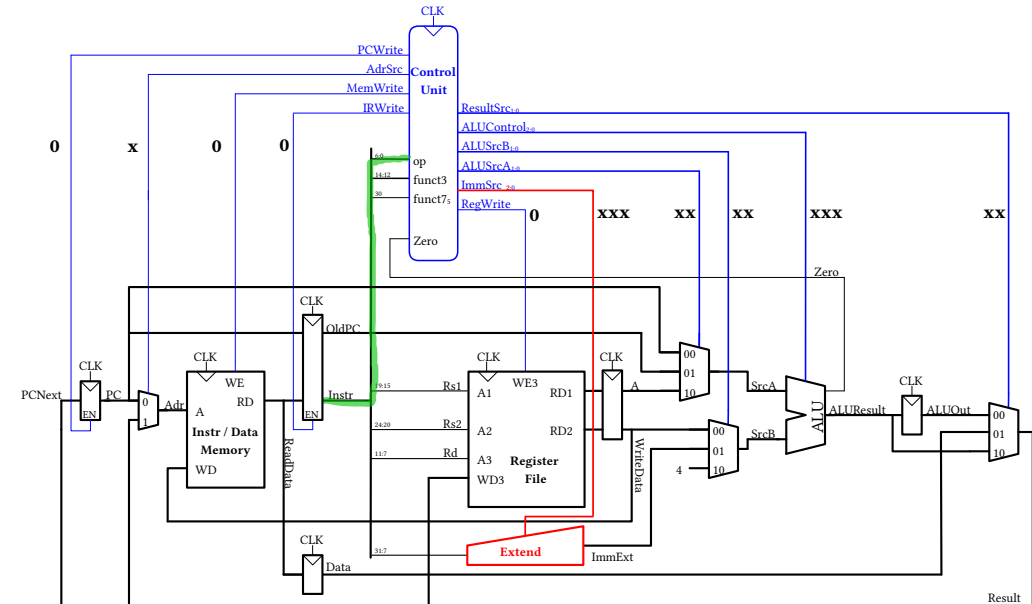
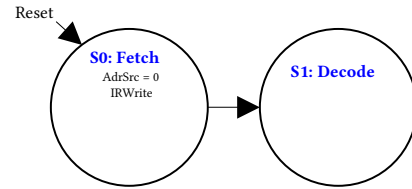
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31	25	24	20	19	15	14	12	11	7	6	0	
funct7		rs2		rs1		funct3		rd		opcode		R-type I-type S-type B-type U-type J-type
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imm _[11:5]		rs2		rs1		funct3		imm _[4:0]		opcode		
imm _[12,10:5]		rs2		rs1		funct3		imm _[4:1,11]		opcode		
imm _[31:12]								rd		opcode		
imm _[20,10:1,11,19:12]								rd		opcode		
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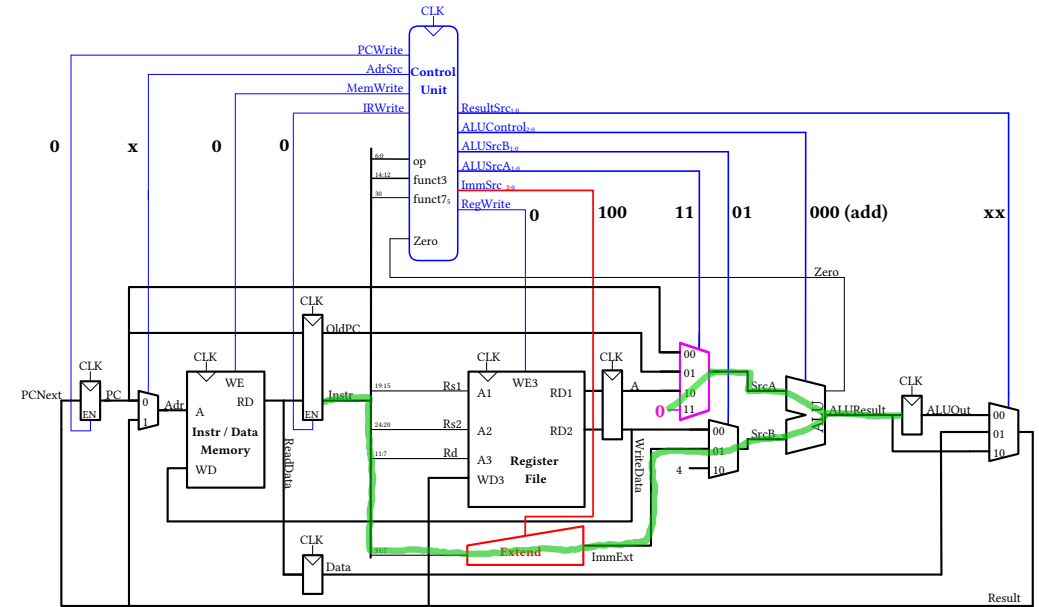
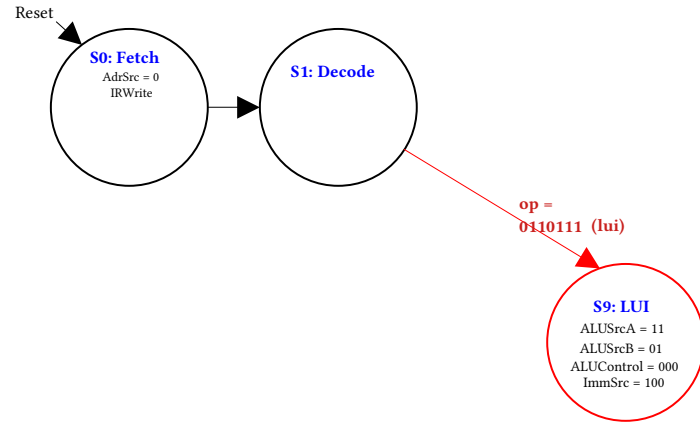
lui: Fetch



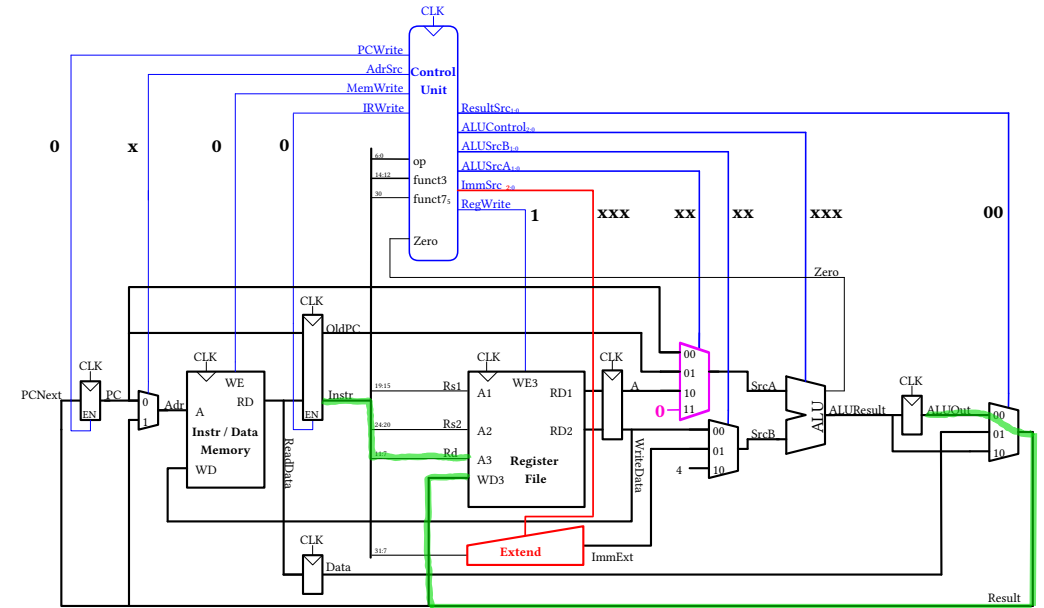
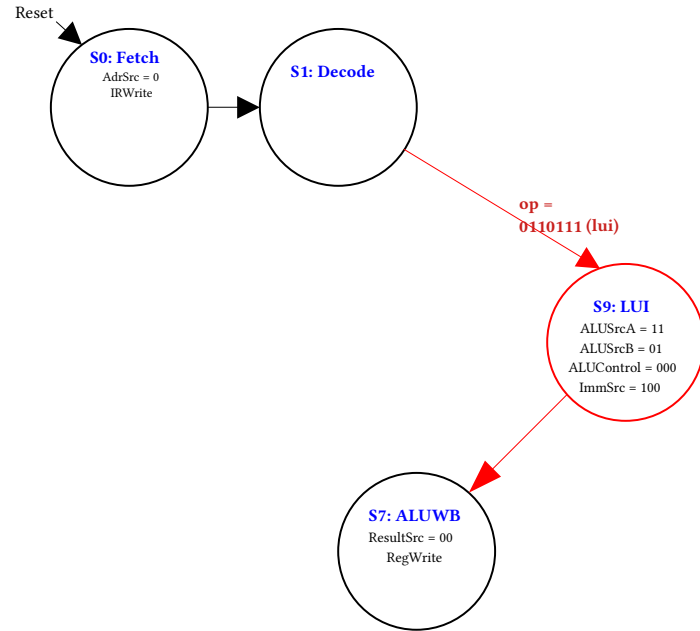
lui: Decode



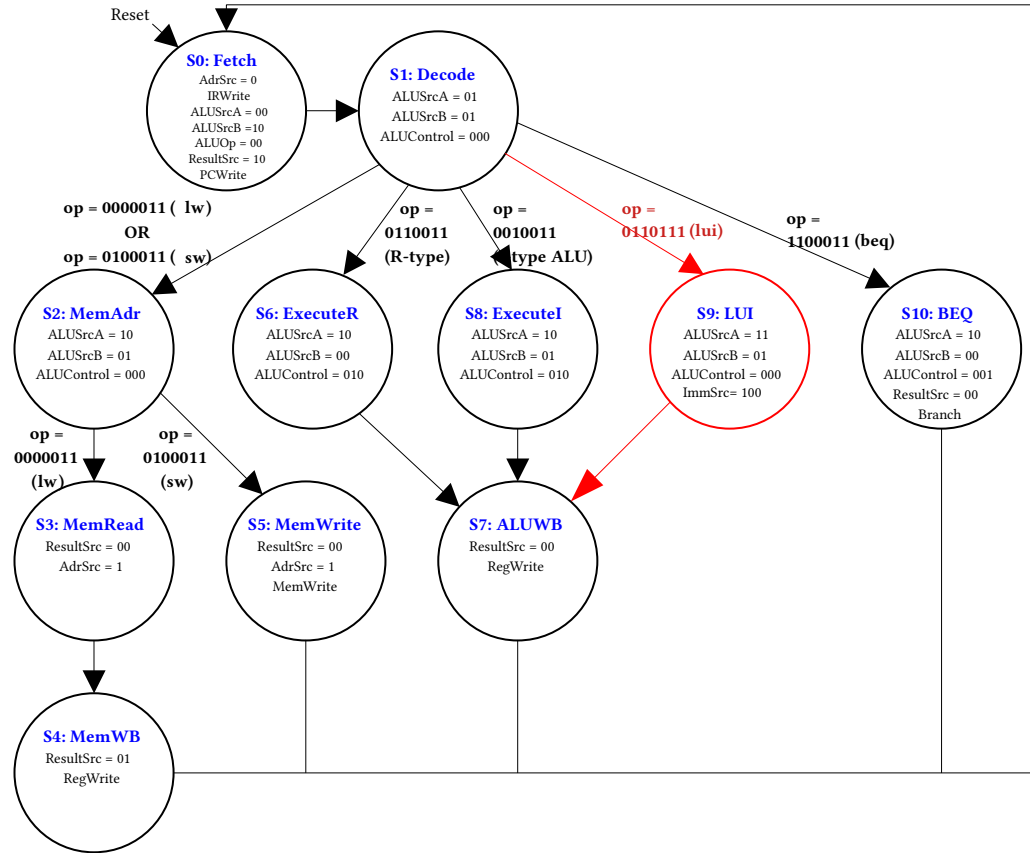
lui: LUI



lui: ALUWB



Komplete FSM



Alternative Implementierung

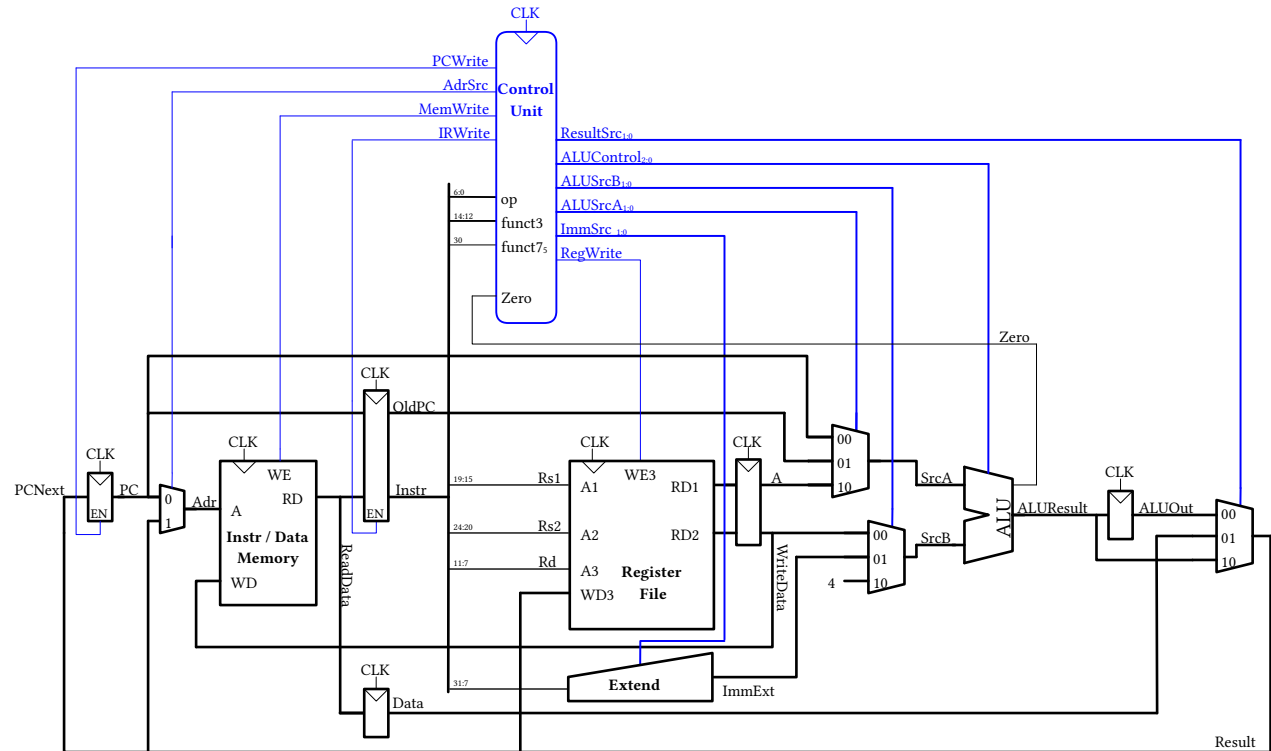


https://sonic-rv.ics.jku.at/example/ics-nex-rv32i-mc/add_immediate

Pfadanalyse

Komponente	Var	Delay
Register clock-to-Q	t_{pcq}	40ps
Register setup	t_{setup}	30ps
Multiplexer	t_{mux}	30ps
ALU	t_{ALU}	120ps
Extend unit	t_{ext}	35ps
Memory read	t_{mem}	200ps

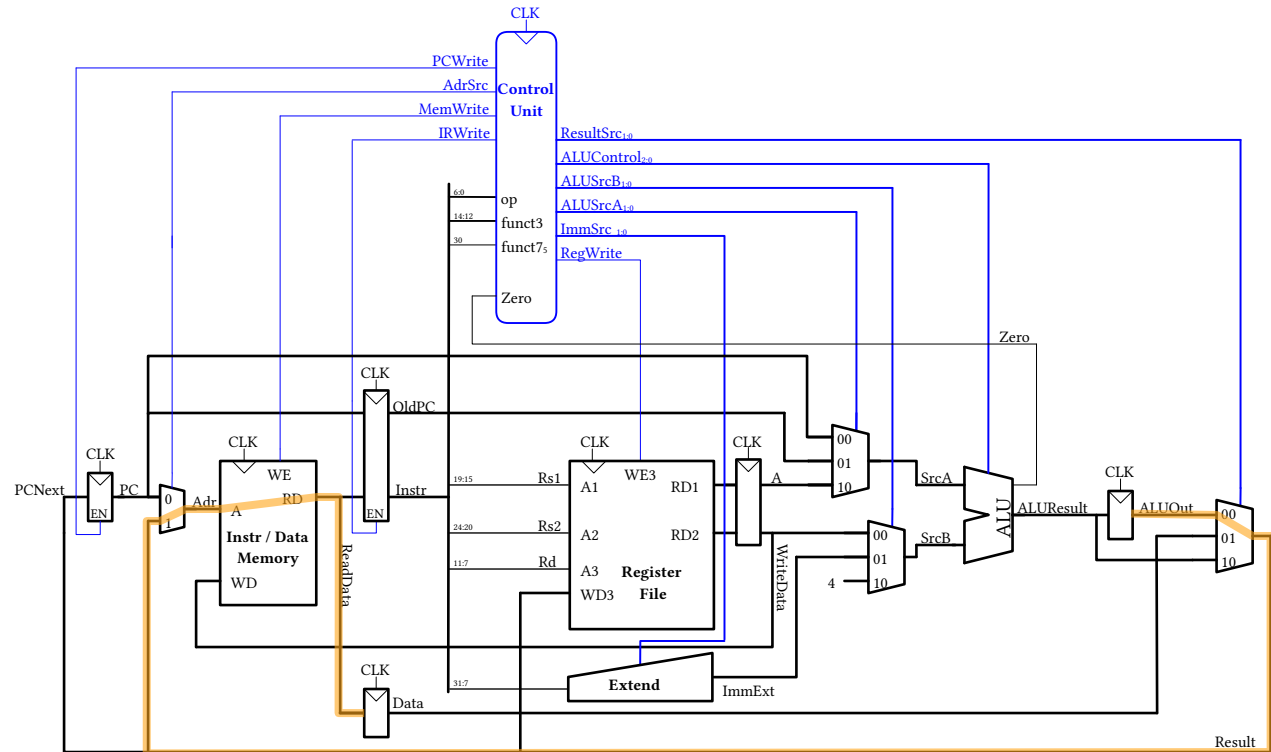
Was ist der längste Pfad?



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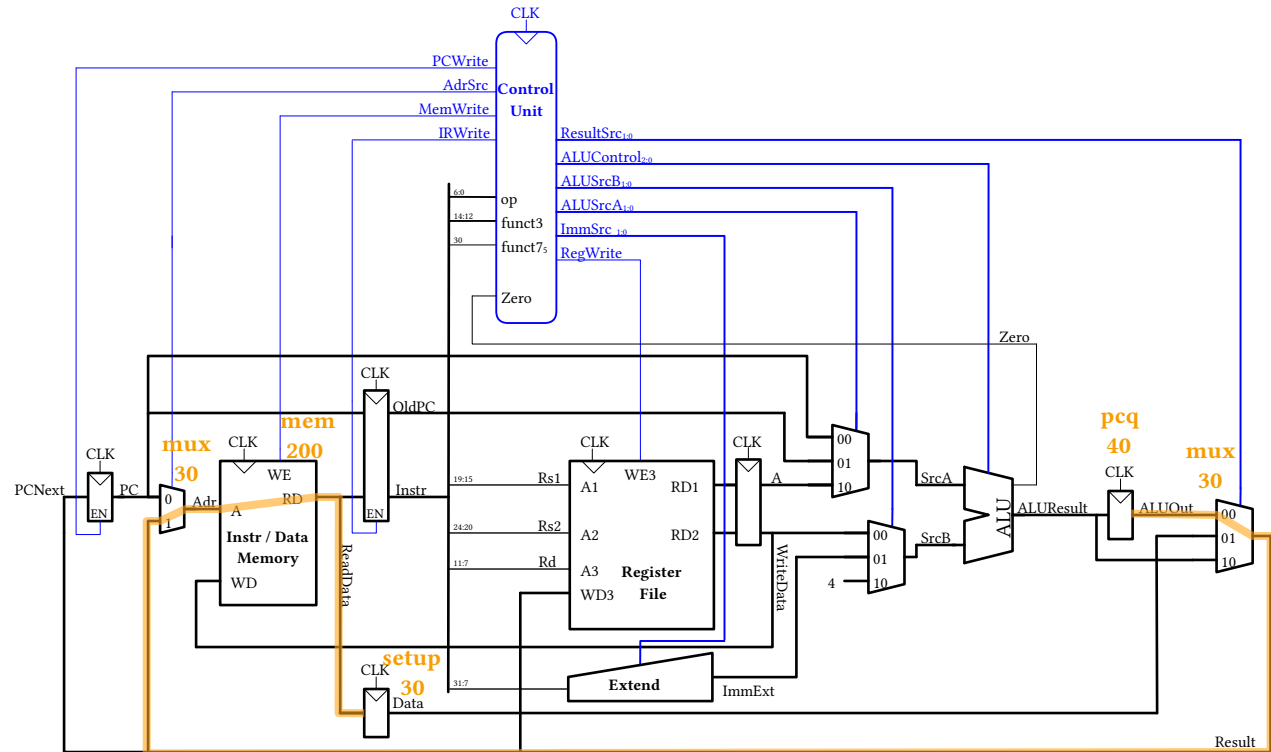
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$$T_c = t_{pcq} + 2t_{mux} + t_{mem} + t_{setup}$$

$$T_c = 40 + 60 + 200 + 30 = 330\text{ps}$$

$$F_c = \frac{1}{T_c} = 3,03 \text{ GHz}$$



DAS WAR'S

Bei Fragen oder Problemen zu mir kommen